

TRI-STAR FALCON (TSF-2A): A NEW AERODYNAMIC DESIGNED UAV WITH HIGH POINT STABILITY AND SPEED FLIGHT MODES.

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Abstract

In this paper we will discuss about a newly designed model of UAV which has been specially made for Vertical Take-off & Vertical Landing (VTOL) and as well as a mode which converts it to a high speed flight mode UAV which has been never been designed yet with a better feasibility factor. The old forms of Quad-rotors have only the VTOL advantage but fail to escape from sight when the enemy falls into the view, this lacking the stealth mode. And apart from mechanical side, it has features like GPS tracking, Live video streaming, High point stability, Stealth modes and more. Features of the UAV, it had fail safe modes like, if the drone is with one damaged motor, then the back or front part actuators can individually bring the drone back to base station acting as the backup and main drive.

Keywords: UAV, Drones, Flight mode, VTOL Mode, Booster Motors

Introduction

The unmanned aerial vehicle (UAV) is a small plane which flies in an autonomous way with only one virtual pilot on board. That does not mean that the plane is completely out of control once it leaves the track. The presence of a human being in the loop of control is essential. But the operator is on a station on the ground, either on ground, or at sea or in a control center. He can intervene at any time, examine the situation and the rules of engagement, and stop the mission if necessary. The concept of an UAV resounds in several research laboratories of any developed country. The first instigators of this type of the projects were the soldiers. Indeed, the armies of many countries prefer to rather send UAV's for the

Advancements

In normal UAVs like the Reaper and Panther it contains the universal flight mode but lacks fail safe modes of landing at high terrain places during emergency thus weakening their part. Now combining both the features brings to an ultimate UAV which proves a lot during the war field. But it's not that simple to design such a model as it need a lot of mechanical factors which falls good during aerodynamic conditions as a small force point at the non-pressure points will lead to damage and crash landing. There have been jet fighters like the Jet Harrier with VTOL and Flight modes but they are for attack. And there have been a lot of designs for Surveillance with both these features but no one has yet brought down to a feasible and marketable application. We have worked on the mechanical design now which stands to be feasible as it being tested on the aerodynamic use as the mechanical design has been tested well. The aerodynamic and mechanical simulation are described here which will prove to its feasibility. More from the mechanical side, it can stop very swiftly during flight modes in the case of emergency. It does not have to take a gliding turn when needed, it just needs to stop at the point and turn at the point itself thus saving the flight time. In brief, in this abstract we'll discuss about some of the mechanical design, and its simulation and analysis. The photos are provided for a well virtual creation and analysis.

Description of the UAV

As you can see in Fig.1, it's in the normal extended flight mode now. In the design, there are two movable wings at the end of which are connected the brushless motor and the tri blade. These wings are connected to the high torque servo motors which are installed in the hull of the drone. These wings stay vertical during the take-off or landing mode. Adjacent the point of brushless motor is there is another servo motor whose shaft is connected to the secondary wing which unfolds during the flight mode. These secondary wings help for gliding in the air thus saving the power of the battery. Now to the back of the drone, we have the booster brushless motor which gives the speed to the UAV. They are also

movable to a resolution of 90 degrees. During VTOL mode Fig.2, it's also in the vertical position, and this feature of having it vertical brings to the fact that it won't restrict the direction of take-off. This will cut down the wind and pause and instability coming in its way.



Figure 1. Flight Mode

Now during the flight mode the main wings rotate 90 degrees to horizontal position and the secondary wings also retract or extend which brings it to the flight mode. The booster motors also rotate to horizontal positions and applying the necessary speed to the glider.



Figure 2. Vertical Lift off and Landing Mode

This UAV as shown in Fig. 3 is 2 meters long and provided with an elevator wing of the scale of 40 m. the profile of this wing is of type NACA 2412 with a cord of 0.236 m, stalled on the fuselage of 3° with a dihedral of 2°. The elevator group of the shape of T reversed is profiled with Naca 0012 and attached to the wings by two beams which

give also the name of an UAV to tristar. These UAV's aspire of a remarkable stability to the no desirable movements of roll. This UAV is designed to have a booster motors which will be embarked behind the fuselage.

Airfoil Wind Tunnel Test

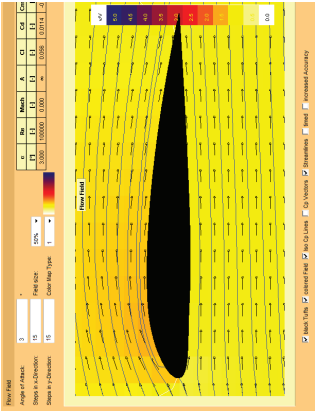


Figure 3 Velocity Ratio Analysis

Several compilations were carried out for various angles of incidence to determine the lift and drag. Each case was carried out at the same speed: 10 m/s. The angle of attack extended from 3 to 18 degrees. It is a parabolic type tendency similar to those found on the planes in general. In particular, one notes the increase in the drag with the increase in the angle of incidence. This curve is presented with drag of the isolated wing which equips the UAV in question and which was obtained by wind tunnel test. The two diagrams follow the same tendency separately which is normal because variation of the drag between the two curves corresponds to the drag of the remainder of the UAV which varies very slightly with the incidence. The current NACA airfoil proves to be having a low turbulence and a better lift factor with the angle of attack set at 3degrees. The velocity ratio is balanced properly. And the pressure generation around the airfoil is distributed symmetrically converging into a stable airfoil design.

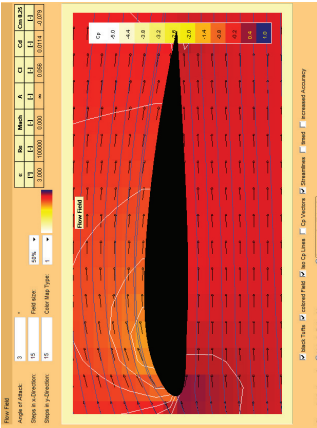


Figure 4 Pressure Co-efficient Analysis

Eppler & NACA Analysis

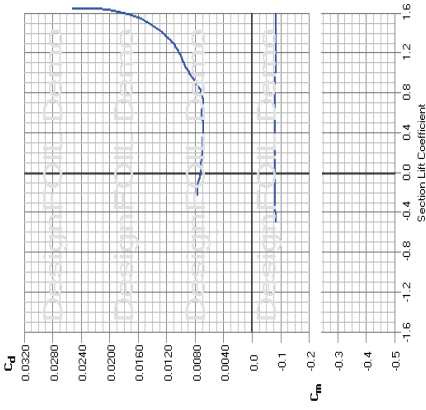


Figure 5 NACA Analysis of Cd vs Cl

An agreement of the two curves is noted in the linear zone which is explained by the fact what it is the wing which contributes in a great proportion to the total lift produced on the plane. The other parts of the plane, in particular the stabilizer contribute slightly towards weak and moderate incidence. However the lift of the isolated wing exceeds that of the plane for the bigger angles of incidence. This is dues with the fact that the Reynolds number of the tests of the isolated wing is much higher than that of the plane tested numerically which leads to higher maximum Cm

for the isolated wing. Numerical calculations show well the phenomenon of unhooking of the UAV which is located at approximately 3 degrees. It should be noted that this incidence which appears is weak only seemingly because it should be mentioned that the reference of measurement of the angle of attack is the base of the fuselage and that the wing is fixed with 4 degrees, flow return is noted on almost the totality of the wing, in particular, it is much attenuated in the central part.

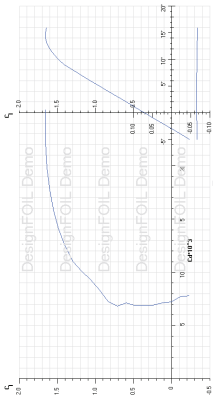


Figure 6 Eppler Analysis of Cd vs Cl

This effect is expected because the wing is not twisted and the great deflections of the air are perceptible in this area. The separation of the flow recorded on the nose of the fuselage still persists with this angle of incidence, flow recorded on the level of the wing root with the fuselage with 15° of incidence.

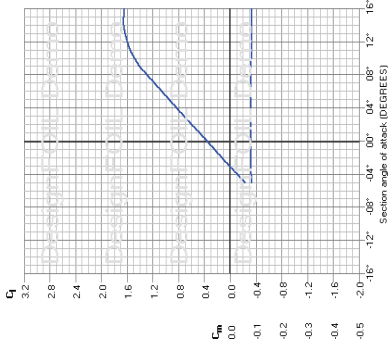


Figure 7 Angle of Attack vs Cm and Cl

Wing Stress & Loading Analysis

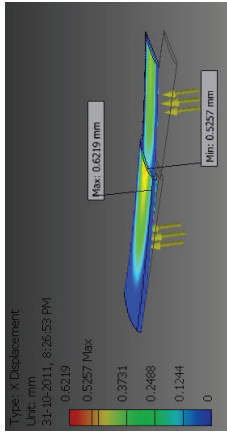


Figure 8 Wing Loading Effect

While both aerodynamic and structural metrics are included in the wing design, the analysis are not explicitly coupled in an aeroelastic sensor, stable deflections of the wing due to air loads are assumed to be small, since large ones will lead to buckling failure. Aerodynamic performance of the wing is studied using the fixed wing stress analysis. Aerodynamic analysis computes the pressure distribution over the wing, as well as the concomitant lift, lift-to-drag ratio (L/D) and the static stability margin. An arc length method based nonlinear deviation tool is used to evaluate the structural performance of the wing and to determine maximum flying velocity that the wing can withstand without buckling or failing (laminate failure with factor of safety = 1.5) under aggressive flight loads.

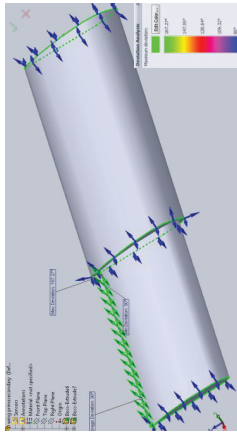


Figure 9 Deviation Analysis

Experimental validation of the numerical procedure used for determining maximum flying velocity has been carried out and can be found elsewhere. An analytical approach based on classical laminate plate theory and strain superposition (initial chord-wise strains due to root airfoil flattening and additional span-wise strains

due to incremental rolling) is used to study the stresses developed in the composite wing during the rolling process for storage with a safety factor of 1.5 is used to check first ply failure of the composite wing to determine minimum safe storage diameter.

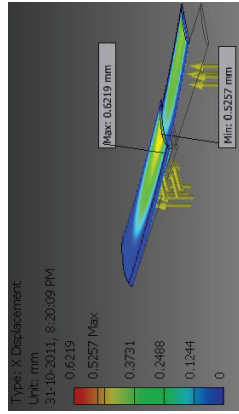


Figure 10 Thrust & Loading Effect

Taking advantage of problem symmetry, we model only one wing half, with the appropriate symmetric boundary conditions at the root airfoil. And appropriate results have been found as the stress analysis proved correct and the wing was able to sustain the pressure at the trailing edge as we rarely see reddened part on the airfoil. Reaching to a corrected figured value achievable to 10m/s or 20m/s.

Components

Onboard controller hardware: Following a minimalistic approach, the central controller board was kept as simple as possible in order to reduce cost and failure rate. It consists of Razor IMU 9 degrees and an AVR Arduino microcontroller. Flight time up to 120 min. (without Payload, 5.8Ah lipo battery)



Figure 11 Arduino 2560

It has very lean design, this controller is very capable due to efficient control algorithms. The central controller board is used to read sensor-data, compute angular velocities and angles in all axes and to run independent control loops for each axis. In addition, the control-outputs are combined to compute a desired speed for each motor, which is then transmitted to the respective motor controller. As piezo gyros suffer a high temperature drift, an inbuilt microcontroller algorithm is used to compensate the sensors' drift before amplifying the outputs. Thus, the highest accuracy can be achieved. All processing is done with a control loop frequency of 8MHz. The main consequence of high frequency control is a low drift rate of the relative angles, as errors arising from time discrete integration are small, and a very stable flight because of very short deadlines in the control loop. Furthermore, the high update rate facilitates FIR filtering sensor data in software without generating big delays. This capability reduces vibrations and shakiness during the flight.

Control System

For TSF, a PID controller was chosen due to its proven capability in similar systems and its ease of implementation. Because the vehicle was a new design, it was immediately apparent that the time to develop, build, test, and refine the design of this vehicle would be substantial. Thus, an effective yet relatively simple control scheme was ideal, because it minimized the time required to solve this important and complex problem. A number of similar applications and vehicles which used a PID controller were found and it was discovered that the performance of this type of controller would be adequate to complete the competition objectives. Additionally, this type of controller allowed returning based on observable behavior. Once the gains were set near the optimal values it was possible to tune the vehicle empirically, which reduced the reliance on the (potentially incomplete) dynamic model and provides a measure of robustness.

In many C or FORTRAN controller, the system input is comprised of three terms which are related to the difference (error) between the current system state and a desired state. The proportional (P) term multiplies the error by a gain value and uses this as an input. This increases the influence of this term

when the current state is further away from the desired state. The integral (I) term uses the integral of the error over time multiplied by a gain value to influence the input. This generally has a gradual effect that reduces steady state error. The derivative (D) term, takes the derivative of the error and multiplies this by a gain value to influence the input based on how the error of the state is changing. This has a dampening effect which decreases oscillations in the system response. TSF uses four PID controllers in each of its roll, pitch, yaw, and height maneuvers.

The onboard controllers are three independent PD loops, one for each rotational axes (roll, pitch and yaw). Angular velocities measured by the gyroscopes and computed relative angles are used as inputs. The angles are derived by integrating the sum of the output of one gyroscope and an external control input for the respective axis. Without an external input signal the calculated integral represents the angle the flying robot has turned in the respective axis. Looking at the closed loop and disregarding measurement noise and integration errors, this means that the robot will always keep its current orientation. The integrated angles can be shifted by an external control input. As a result, the robot's orientation changes.

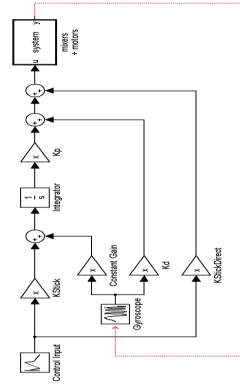


Figure 12

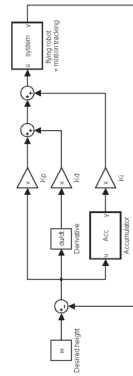


Figure 13

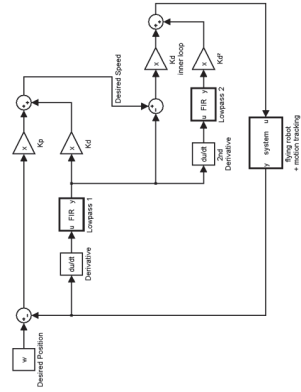


Figure 14 Basic structure of the onboard control-loops.

proportionally to the input. Its movements are controlled by steering it to a certain orientation and keeping this orientation for a certain time. Due to measurement noise and discrete integration the integrated angles drift about ± 3 degrees per minute. However, this drift can be easily compensated by a human pilot or an autonomous external position control such as a motion capture system. The controller implementations have been optimized for shortest possible execution time and robustness in almost every flight situation. The loop is interrupt triggered, which enables stable time constants for integration and filtering. By using the AVR's internal ADCs at a high sampling rate, fixpoint arithmetics only, runtime optimized FIR filter implementations and interrupt driven communication to update the motor speeds, we achieved a system running at a control frequency of 8MHz.

All controller parameters have been set empirically and optimized experimentally over several months.

During this period both, hardware and software, have been optimized as far as possible. The result is a very reliable hardware revision of the central controller board, as well as a set of controller parameters capable of reliable control during slow movements as well as during fast maneuvers, even including loops where the robot is inverted for short periods.

Sensorless Brushless Controllers: Our robot uses brushless DC motors. Unlike brushed DC motors, brushless motors are commutated electronically rather than electromechanically. The common

brushless motors use three phases. Current is always floating through two of these phases, while the third phase floats free and is used to measure the angle of the motor. Then the controllers commutate electronically with a three phase H-bridge. The time when the third phase crosses $V_{dd}/2$ is called zero-crossing point and triggers the next commutation step after a certain time. The three phases are driven in a semi-sine mode, where the phase difference between any pair of phases is always 120 degree. Most commercially available sensorless brushless controllers are sold in the model aircraft market. These are controlled using servo impulses with a 50Hz update rate. To achieve better stability in every situation, we wish to reduce the deadline in the control-loop from the gyro measurement to the torque change of the motor. We designed the system for a control-loop frequency of 8MHz. The microcontroller used on the brushless controller is an Arduino 2560. This microcontroller was selected because it offers all required hardware features like timers, PWM generators.

A challenge with sensorless commutation is the necessary minimum speed to detect the position of the motor. The start up is done open loop in a stepping mode to accelerate the motor. The loop is closed afterwards to control the commutation. As soon as the motor runs in the closed loop mode the current is regulated by the PWM that is commanded by the central controller board. The commutation loop maintains a synchronization between the motor and the electrical commutation. In the worst case, the PWM is updated latest on every commutation. Our motors run between 2000 and 40000 rpm. The motors have two electrical commutations per mechanical commutation. The other advantage of our brushless controllers is the optimization for the low-rpm optimized brushless outrunner motors.

Brushless Motors and Rotors: The brushless outrunner motors used in our flying robot are a special design for high rpm applications.

Autonomous Flight

We implement autonomous flight by using an external sensor system (i.e. motion capture system) to compute the position, height and yaw for the robot. The sensor system can be GPS or

DGPS for outdoor applications, or any kind of indoor tracking systems like a sensor node network, an ultrasonic position measurement or an optical motion tracker.

Cartesian coordinates of each marker, the robot's position and orientation can be determined using simple geometry. The markers' positions are transmitted via a TCP/IP-Link to a computer running the position control algorithms. After identifying the markers by mapping them to a model of the robot, the robot's orientation and position is calculated and provided as real-time input to the controllers. The update frequency of the position controllers is set and limited to 50Hz due to the limitations of the R/C transmitter used for sending commands to the flying robot. The performance and stability of the onboard electronics make this external control loop frequency of 50Hz adequate for achieving stable flight.

However, we had to modify the internal electronics using another Arduino microcontroller to connect it to the laptop. The protocol of the serial interface allows us to select a source independently for each of the channels. The source can either be the joystick for human control or the PC-software. This system has a user interface for developing the position controllers which enables debugging, testing and optimization step by step.

Position control: The laptop receives the datastream from the motion tracking system and outputs data to the transmitter. There are four independent controllers running on the laptop computer. They are implemented using a customized C++ software module. The control loops are timer triggered to enable a precise 50Hz update rate. The Yaw-Controller was implemented as a PD loop. Inputs for the controller are the measured yaw angle, its FIR lowpass filtered derivative, and the desired yaw angle (heading).

Navigation

We have implemented a scan matching algorithm in order to obtain position information on the horizontal plane throughout the competition course. The scan matcher approximates the rigid

body transformation between adjacent discrete time poses represented by the output of a single planar laser scanner. Once the transformation is known between neighboring poses, determining our global position is accomplished by applying the history of transformations to our known initial pose. Furthermore, the robot uses these transformations to build an occupancy-grid map from the raw laser scans in order to perform higher level tasks such as collision avoidance and goal oriented path-planning.

Decision Making

The navigation has several components after the generation of the map by the scan matcher. The vehicle must utilize the laser scanner and camera inputs in conjunction, weighing the information which each provides. Additionally, it keeps track of its mission state and prioritizes the inputs based on the current objective.

The camera is operated through the use of OpenCV, a library of image processing algorithms. In order to increase the flexibility of the image processing algorithm, a program was written which sets the number of frames per second of video capture along with the total number of stored frames. This flexibility is crucial in order to minimize the delay from the cameras and to maximize the efficiency of the object detection algorithm. Individual frames from the stream of each camera are analyzed to check for recognizable objects.

Object Detection

After acquiring the captured image from the camera using the algorithm mentioned above, TSF must be able to distinguish whether or not each image contains an object of interest. Our system utilizes OpenCV's method to accomplish this. It uses thousands of images designated as "positive" or "negative" images to learn the distinguishing features of an object of interest. For good results, many positive and negative images need to be processed by the training program.

Testing and Optimization

Once the program has been trained to recognize an image, OpenCV tests recognition percentage by overlaying positive and negative images. The training performance program is then used to determine how many positive images the computer is able to detect. If training is done correctly, the computer should be able to detect and point out over 99% of positive images. Once a sign has been recognized, the image processing code sends a flag to the navigation software for a proper response.

OPERATIONS

Man/Machine Interface

TSF's custom circuit board features eight LEDs that inform the team what the vehicle is doing at all times. With these LEDs, TSF members are able to determine if the motors are armed, if the control loops are running, and if the onboard Atmega processor is functioning properly. It is also possible to program the LEDs to check for specific problems during troubleshooting.

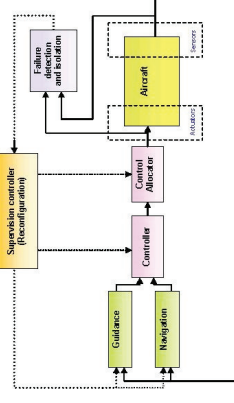


Figure 15
Fault tolerant system overview

Modeling and Simulation

TSF was modeled in Solidworks to determine its physical characteristics such as moments of inertia and center of mass. This data, along with motor thrust values were used in Simulink to predict vehicle response. Proper PID gain values were then empirically determined and fine-tuned from this simulation.

Flight Testing and Safety

Throughout the testing phases of our vehicle we have implemented the use of a tethering system to keep both the vehicle and team members safe. Carbon fiber propeller guards prevent the propellers from shattering on impact with a solid surface. Our system provides several fail-safe's to prevent vehicle damage and personal injury. In the case of an emergency, our motors can be killed in three independent ways: by joystick control, by a command issued from the ground station.

Vehicle Status



Figure 16 Ground station GUI

TSF vehicle status can be monitored through the graphical user interface (GUI). The GUI reports critical flight data such as vehicle altitude, battery life, and communication signal strength as well as the mission state. With this information, we can predict what issues could possibly arise allowing us to better react to a situation and reduce risk as much as possible.

Shock/Vibration Isolation

The outer ring of the airframe connects the motor attachment points and reduces the vibration generated at each location. Additionally, soft spacers have been installed between the electronics board and the airframe, protecting electronic components from unintentional vibration.

Conclusion

The present study was undertaken to observe the aerodynamic performances of an UAV by using a CFD computer code. This concept of design of flying machines is judged like an alternative to the wind tunnel tests for a possible reduction of the design cost. This work brought moreover, certain important aerodynamic corrections around this UAV such as the aerodynamic stability of the nose of the fuselage and the setting of the bases of fastener of the beams on the level of the under-surface of the wings with the proper stress analysis done.

Several change and simplification were made in the modeling process, including the use of electric brushless motors instead of reciprocating, the constant mass, counter-rotating propeller to eliminate rotating-mass effects, the removal of aileron control, etc. These changes were made in a way that the flight characteristics would not change much. The simulations were conducted several times performing a take-off, hover, cruise and landing maneuver. Softwares like Solidworks, Autodesk Inventor, Design Foil and many others have contributed a lot for the analysis and simulations done in this paper. It's been a successful path along.

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